RMIT UNIVERSITY

Full Stack Development S1 2026

Assessment Task 3: Individual Programming Test

Business Case: Hotel

KINDLY READ FIRST

1. You are allowed **15 minutes** to read this Business Case Brief.
2. You must not spend more than **1 hour** to complete this test.
3. Accept the invite to A3 https://classroom.github.com/a/2vNbn3g9 which will create a private student repository with this format **a3-individual-programming-test-<your_github_username>** under **rmit-fsd-2026-s1** organisation.
4. Submit the URL of this repository via the Assignment 3 page in Canvas. **Failure to submit the repository URL within the practical session attended will result in a score of zero for the assessment, even if the repository exists**.
5. Any changes to code in the repository outside the hours of the practical attended will result in **a score of zero for the assessment.**
6. Only the first attempt made in the first practical will be marked. **Subsequent attempts** made in different practical sessions will be **ignored**.
7. Create a React TypeScript application. **ReactJS** applications will receive a **zero**.
8. Please **shut down all the messaging apps** on your laptop. Do not open any Gen AI programs.

You are tasked with developing a dynamic and interactive hotel room pricing calculator for XYZ Hotels, a luxury hotel chain. The hotel pricing calculator should be built as a **client-side** React application, designed to provide interactive user experience. **No use of databases is required**.

The form will enable potential customers to calculate their room rates, providing detailed breakdowns of the total cost based on room type, location, and stay details. This form will need to communicate with an existing REST API to dynamically fetch and calculate the room's pricing based on user inputs.

Hotel Room Options

Option Type	**Available Choices**
Room Type	Standard, Deluxe, Suite, Presidential Suite
City	Jakarta, Denpasar, Singapore, Sydney, Tokyo
Number of People	A positive number representing the number of guests
Has View	Boolean indicating if the room has a special view
Length of Stay	A positive number representing the number of days
Season	peak, shoulder, off

API Information

The base URL for the hotel pricing calculator API is: https://a3-hotel.onrender.com
Documentation is available at: https://a3-hotel.onrender.com

The "calculate pricing" endpoint is [POST] https://a3-hotel.onrender.com/calculate
The “cities” endpoint is [GET] https://a3-hotel.onrender.com/cities

Tasks

Task 1 (10 marks)

Your first task is to build the user interface (*no API calls yet*) as outlined in the brief above, implementing fields for all the fields outlined above.

The user interface for the pricing calculator application should also include features like a header and a footer with some appropriate content, in addition to the form itself.

The form should have a two-column layout, using CSS flexbox or grid, with half of the options on the left-hand side, and the other half on the right-hand side. The "calculate" button should sit at the bottom of the form and take up the full width of the page. You are allowed to use libraries such as Bootstrap, Material UI, Chakra UI, or any relevant styling module. Please ensure that the form is styled professionally.

Please ensure that every input is a proper HTML5 input field (input, select, etc.) with the name property of the field set to the corresponding value to match the API. For example, if the field in the API is "roomType" then the input should be

```
<input name="roomType" ...rest of input />
```

Task 2 (4 marks)

Implement client-side validation for each form input field. These validation checks should occur when the customer clicks the "Calculate" button:

- All fields are compulsory
- Room Type must be one of: Standard, Deluxe, Suite, Presidential Suite
- City must be one of: Jakarta, Denpasar, Singapore, Sydney, Tokyo
- Number of People must be a positive number
- Has View must be a Boolean value
- Length of Stay must be a positive number
- Season must be one of: peak, shoulder, off

Any errors should be displayed within the form, positioned above or below the corresponding field. For example, if the user enters an invalid room type, the error message should appear above or below the room type input field.

Write your own validation code. The **use of validation libraries is not permitted**.

Task 3

Task 3a (2 marks)

Using Postman or similar, please send POST request to call the calculate endpoint while passing the following payload in the request's body:

```
{
  "roomType": "Standard",
  "city": "Tokyo",
  "numberOfPeople":2,
  "hasView": false,
  "lengthOfStay": 3,
  "season": "shoulder"
}
```

Retrieve the API result in JSON format. Save this output in file named *output.json* and upload it on GitHub repo, as proof that you can use the API.

Task 3b (8 marks)

Using the API provided above, when the customer submits the form, make an API call to calculate the room price based on the selected options.

Display the calculated results to the user, including:

- Total Price
- Detailed breakdown showing:
    - Base Price
    - City Multiplier
    - View Multiplier
    - People Adjustment
    - Season Multiplier
    - Stay Discount
    - Subtotal
    - Discount Amount

Ensure proper handling of any errors received from the API and manage loading states effectively.

For example, display a loading spinner while the API calculates the price.

The calculated total price should also be displayed in the header of the application. You must use React Context to accomplish this.

Use of **3rd party state libraries or Redux is not allowed**.

Task 4 (6 marks)

Write one substantial unit test that makes sense for the scenario mentioned above. Please write code comments to explain the purpose of your unit test.

END OF ASSESSMENT